ThickSense: submitted project summary, iDEATE 2026

Text of the summary as submitted by the team. Layout added for this page.

Here is a short description of our project.

Team Name: CATS (Community Action Technological Solutions)
Project Description:
ThickSense is a next-generation, AI-integrated smart spoon that tells you whether a thickened drink is safe to swallow in three seconds. Powered by sine-wave vibration technology, it works in three steps: dip, press, check. The spoon vibrates in the drink, a sensor reads how much the liquid dampens it, and the display shows the viscosity value, then Too Thin, Just Right or Too Thick. Its machine learning companion app tracks every result, learns each patient's preferred thickness, and gives doctors, therapists and caregivers real-time insights.

Our beachhead market is Singapore's nursing homes and hospices, where dysphagia (difficulty swallowing) affects roughly half of residents. Spoons are sold directly to care homes, families caring for elderly parents and elderlies themselves. Reusable, and targeting $15 against $40 syringe kits and $200 viscometers, ThickSense is an unbeatable value proposition: lower choking and dehydration risk, less waste, and certainty in every cup.